MULTI-CLOUD FDE STACK #01

MULTI-CLOUD FDE STACK · ISSUE 01 OF 08

One architecture, three clouds.

The app is portable. Identity, networking and secrets are not.

01

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Swipe →

The service map

NEED	AWS	GCP	AZURE
Kubernetes	EKS	GKE	AKS
Objects	S3	Cloud Storage	Blob Storage
PostgreSQL	RDS	Cloud SQL	Azure DB for PG
Secrets	Secrets Manager	Secret Manager	Key Vault
Queues	SQS	Pub/Sub	Service Bus
Identity	IAM	Cloud IAM	Entra ID

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What ports and what does not

PORTABLE

- Containers and Helm charts
- PostgreSQL schema + pgvector
- OpenTelemetry instrumentation
- Application code

PER CLOUD

- Workload identity
- Private networking
- Secrets wiring
- Queue delivery semantics

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Where renames break

01 Networks

A GCP VPC is global. AWS VPCs and Azure VNets are regional.

02 Queues

Ordering, dedup and retry rules differ per service.

03 Identity

IAM roles, service accounts and managed identities behave differently.

04 Postgres extensions

Azure needs vector allowlisted before CREATE EXTENSION.

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THE HONEST ANSWER

“It runs anywhere” needs a tested Terraform module per cloud.

Say which clouds you have actually deployed, and what a port costs in weeks.

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Before you promise a cloud

- [x] Mapping table with differences per row
- [x] Terraform module per cloud
- [x] Workload identity tested
- [x] Private endpoints tested
- [x] Deploy rehearsed in their account

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KEY TAKEAWAY

Keep the app neutral. Make the cloud edges explicit.

→ Map every managed service, row by row

→ Call out behaviour, not just names

→ Only promise clouds you have tested

YOUR TURN

Which cloud service surprised you most when you ported a deployment?

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Next in the series: #02 Identity is the first integration

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